

Emerging Trends in Photonics

A patent-landscape cartography of eight fast-growing domains

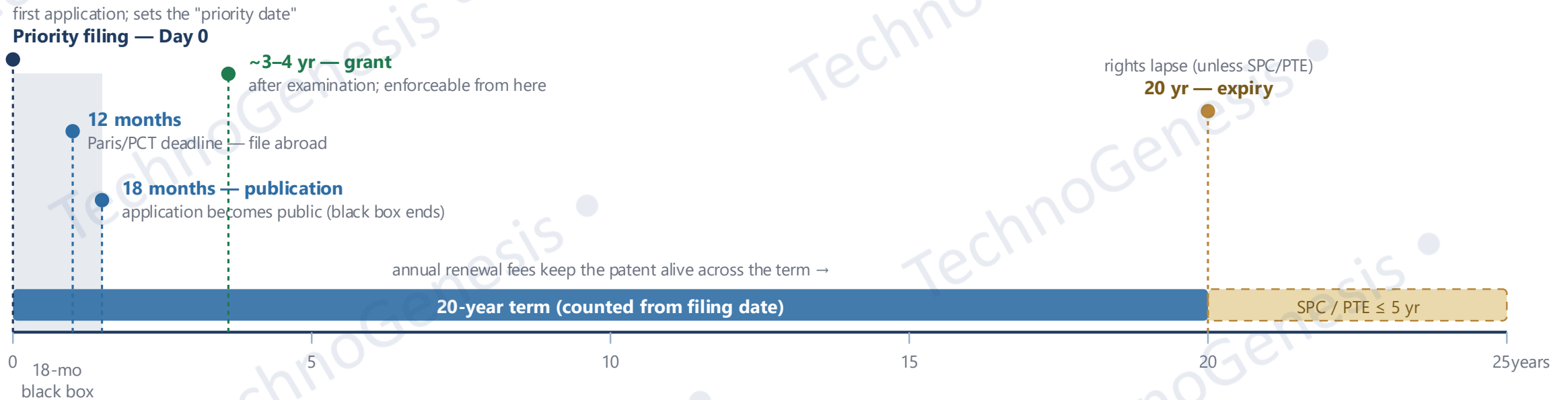
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Keynote · GPOL 2026 · Paris · 20–21 July 2026 | Preliminary, indicative figures

The lifetime of a patent

From priority filing to expiry — and why priority year $\neq$ publication year



Priority year $\approx$ when the invention was made · Publication/grant lag 1.5–4+ years · Time runs from the filing date, not grant

THREE ROUTES TO PROTECTION

- **National** – file office by office (US-USPTO, CN-CNIPA, KR-KIPO).
- **Regional** –EPO (EP) ~ covers 39 states, (UP) ~ covers 18 states since 2024 .
- **International** – PCT (WO); one filing reserves entry to ~ 150 countries for 30/31 months before national entry-> **Is a route – no protection until national entry and grant!**

RIGHTS & PROTECTION

- **Territorial** -> protects only where granted
- **Claims define the legal scope** -> scope of a granted patent can never be broader than the application as filled.
- **publication** – when rights (i.e provisional protection) begins.
- **Grant** when enforceable protection begins.

PRIORITY YEAR = THE INVENTION MOMENT

- **Earliest priority year = when the invention was made**
- **Publication & Grant lag** 1.5-4+years; long procedure which can be longer in some jurisdictions (e.g. postpone examination in Germany up to 7 years)

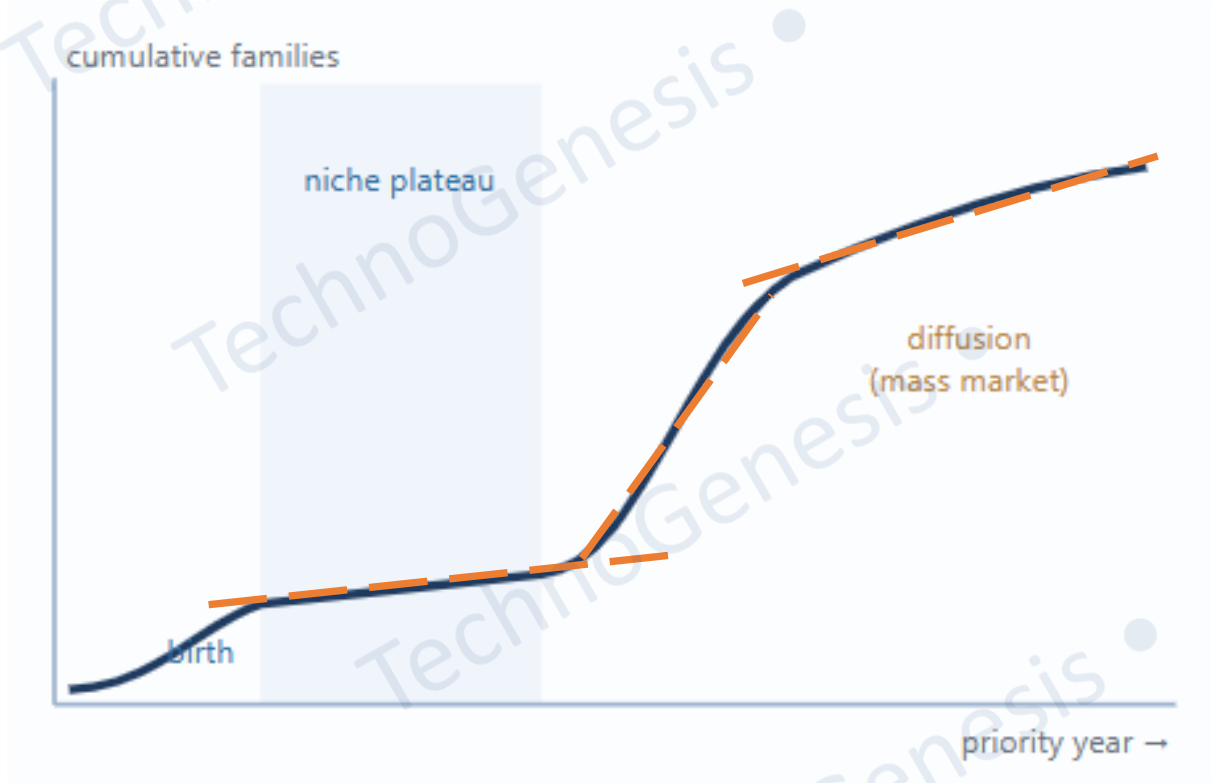
Methodology

- **Patent families analysis** – Family as a unit, plot versus priority year.
- **Combination of IPC/CPC classification codes** (Technology) & **Keywords** - (Title/Abstract/claims). Codes bound the domain, keywords refine them and catch emerging technologies that classification misses. (e.g. hollow –core: G02B 6/02328, OCT: A61B5/0066). One patent application can carry several codes

The 18-month black box & even more!

- **Black box**– Application stay secret ~18 months after earliest priority -> 18 last months can't be considered for an analysis
- **International routes** – EP and WO (30/31 months to know)
- **Volume ≠ value like publication ≠ grant**; counts differ by source & scope.

To take **away**: a count measures activity, not rights or value.



THE LOGISTIC S-CURVE

- **Emergence** -> steep adoption (invasion) -> **saturation** (consolidation)
- **Natural law of how technology spreads** (Bejan & Lorent, 2011)*

*Bejan, A. & Lorente, S. (2011), "The constructal law origin of the logistics S curve," *Journal of Applied Physics*, 110, 024901.

THE TRAP

- A long flat middle can be a **niche-restricted phase**, not maturity.
- Read too early, can't give safe conclusions.

An S-curve is a hypothesis about diffusion-not a verdict on maturity.

Hollow-core fibre

Studied in Minesoft Origin

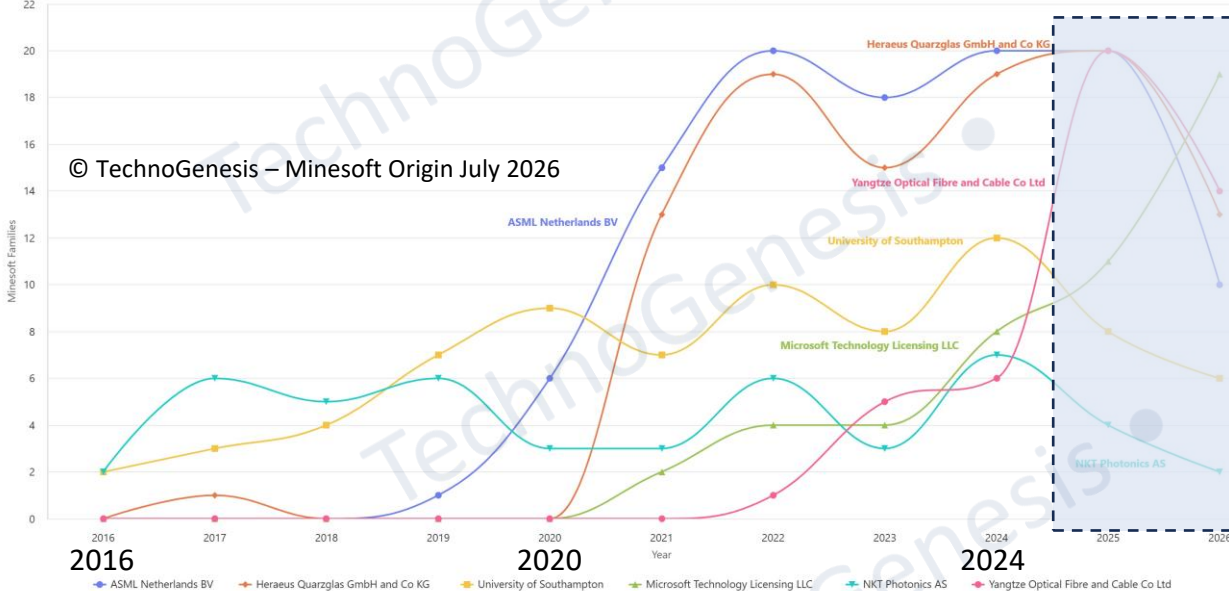
Number of priority application per priority year for the period 1994 to 2026

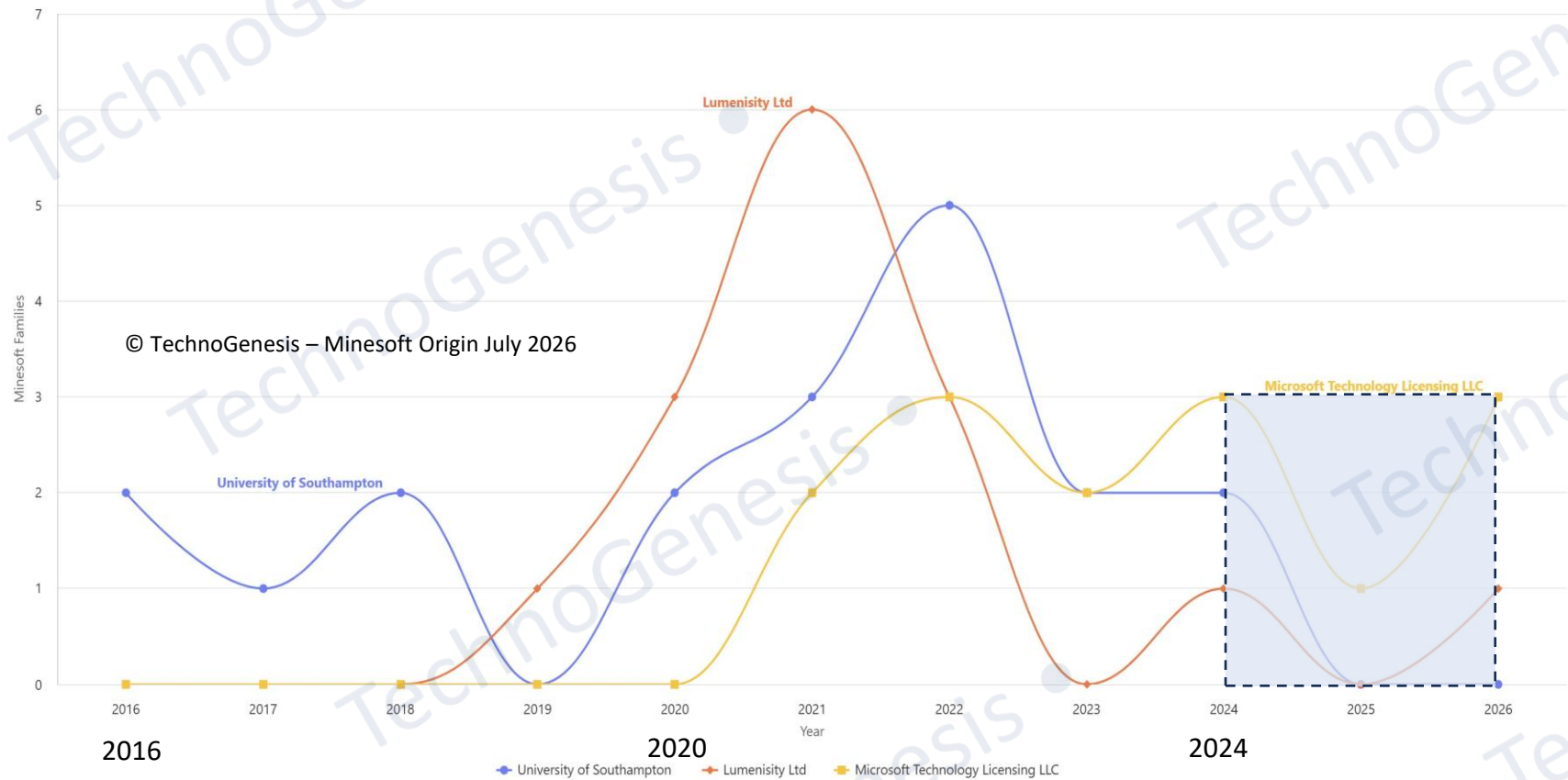


- **Not one S-curve but two waves:** – Family as a unit, plot versus priority year -> 2000-2004 (Photonic bandgap era), long plateau until 2013 (niche phase), 2014 pulled by AI data-centre latency.
- **No plateau reached until now:** – The domain is still open and white spaces remains (manufacturing, splicing and cabling process).

ASML: High-power beam delivery
Heraeus Quarzglas: Silica performer supplier
University of Southampton/Luminesity: Academic origin of NANF/DNANF
Microsoft Technology Licensing: Lumenisity acquisition
NKT Photonics: speciality Fibre
Yangtze Optical Fibre & Cable: China's entry

Top 6 Assignees for the period 2015 to 2026





University of Southampton -> Spin out **Lumenisity** -> Acquired by **Microsoft** December 2022

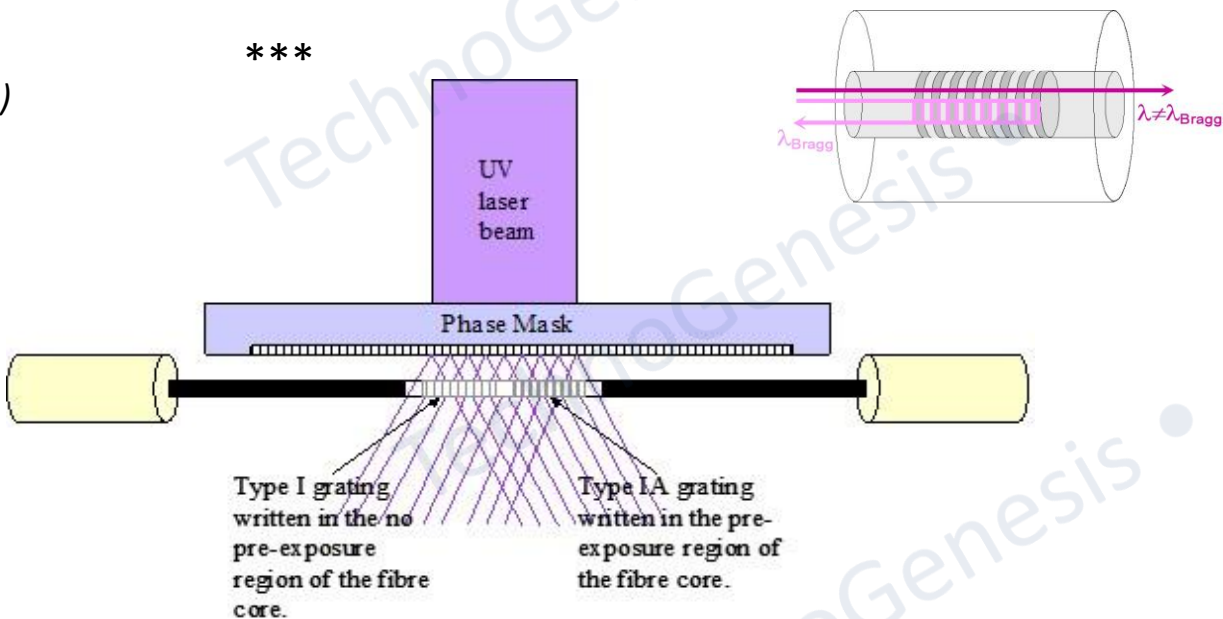
A priority-year S-curve tells you **when** something was invented — not **when**, or **where**, it will matter.

Technology	Early restricted application	Later Growth & Diffusion
Aerospace insulation (Airbus)	Aircraft thermal control	Building envelopes *
Fibre Bragg Gratings	Motorsport (F1) & Aerospace measuring strain/temperature	SHM: bridges, dams, pipelines, wind turbines, medical equipment **
Hollow-core fibre	High power delivery, gas sensing, gyroscopes	AI data-centre & HFT low-latency links
Adaptive optics	Military / astronomy (telescopes)	Ophthalmology, microscopy, free-space optics
LiDAR	Space / Military	Automotive & consumer

* (Accepted Invited talk at 16th Constructal Law Conf, 12-15Oct 2026, Paris)
Project collaboration between TehcnoGenesis & PSL University, Paris

** (FBG medical application) Ren, H. (2020). *Flexible Robotics in Medicine: A Design Journey of Motion Generation Mechanisms and Biorobotic System Development*. Academic Press.
<https://doi.org/10.1016/C2018-0-00607-6>

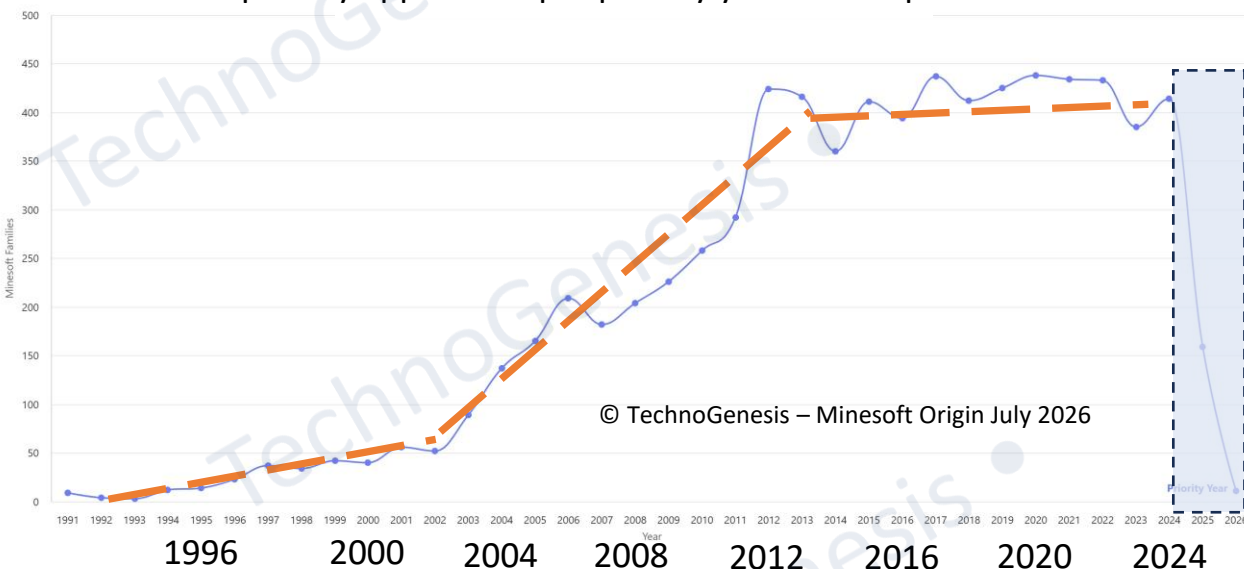
*** (image taken from) Mexis, M. (2004). *Fiber Bragg Grating Sensing Systems*, MSc Thesis, Aston University.



Biophotonics: Optical Coherence Tomography (OCT) & Wearable Optical Biosensing

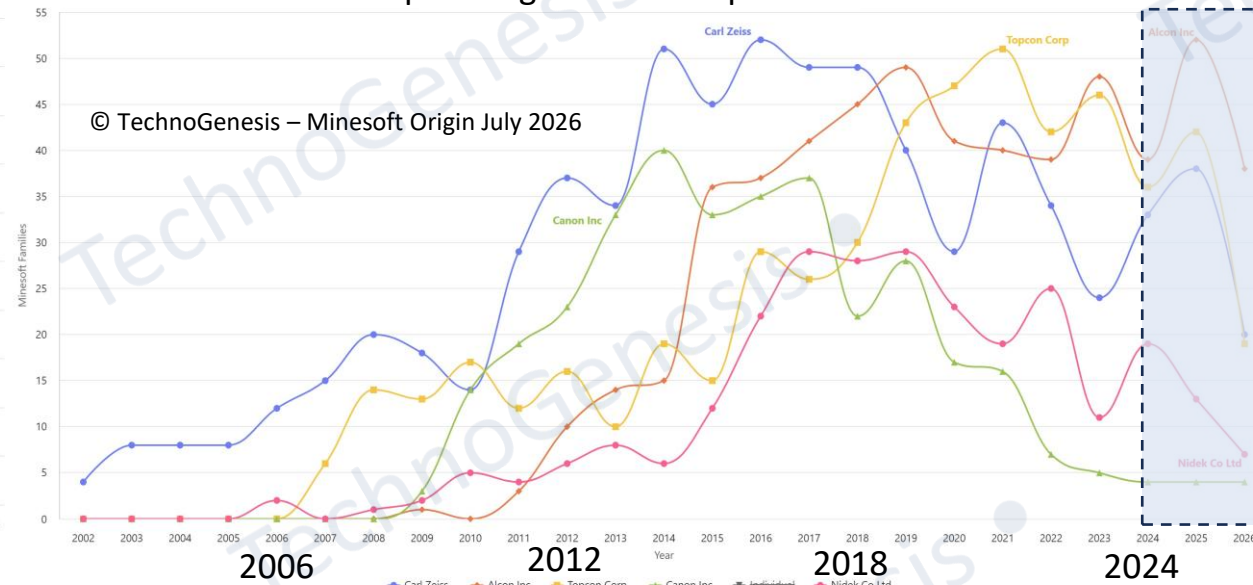
Studied in Minesoft Origin

Number of priority application per priority year for the period 1990 to 2026



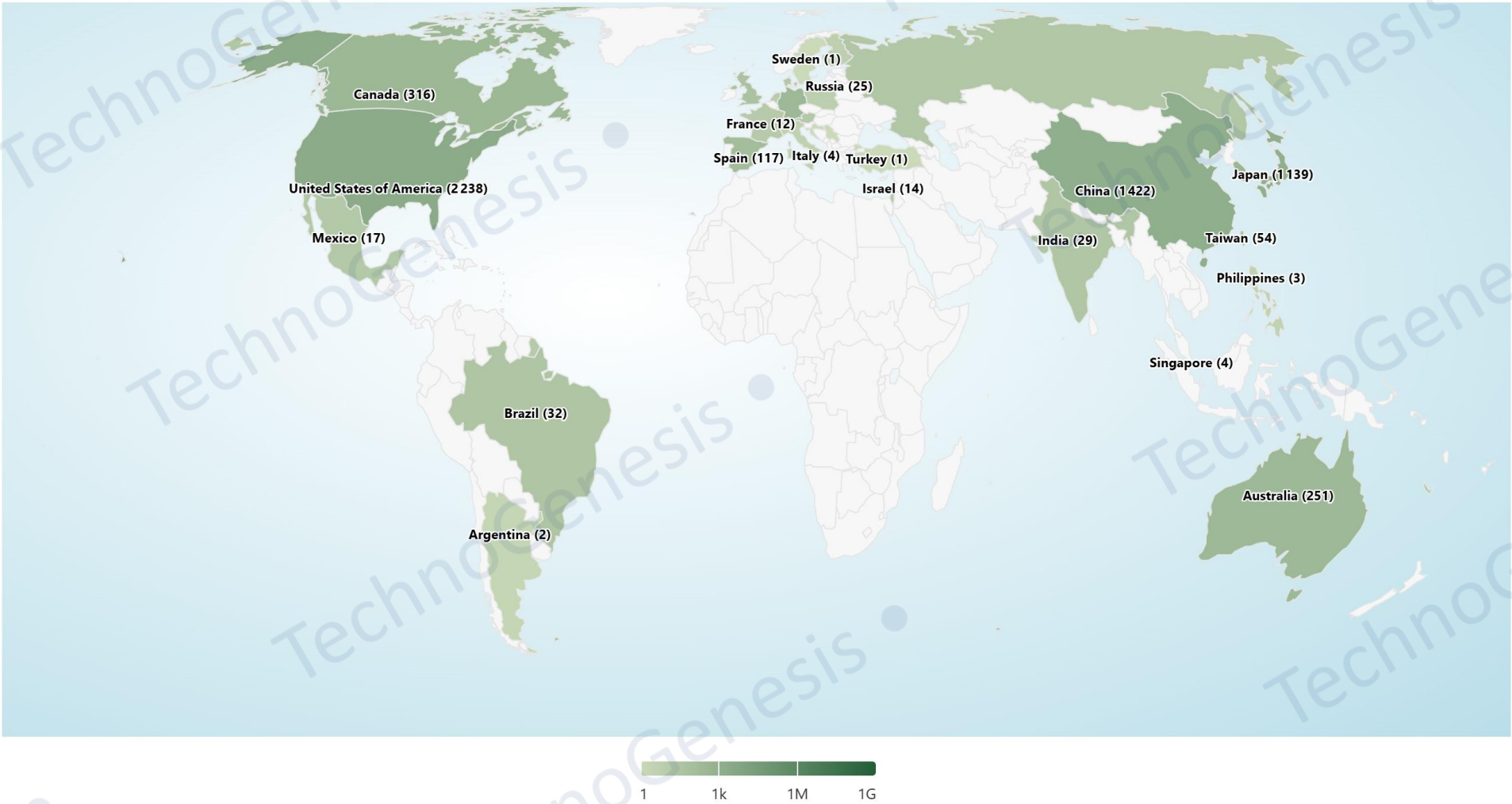
A **mature** domain: steep growth to ~2012, then a flat plateau of ~400 families/yr — saturation.

Top 5 Assignees for the period 2001 to 2026



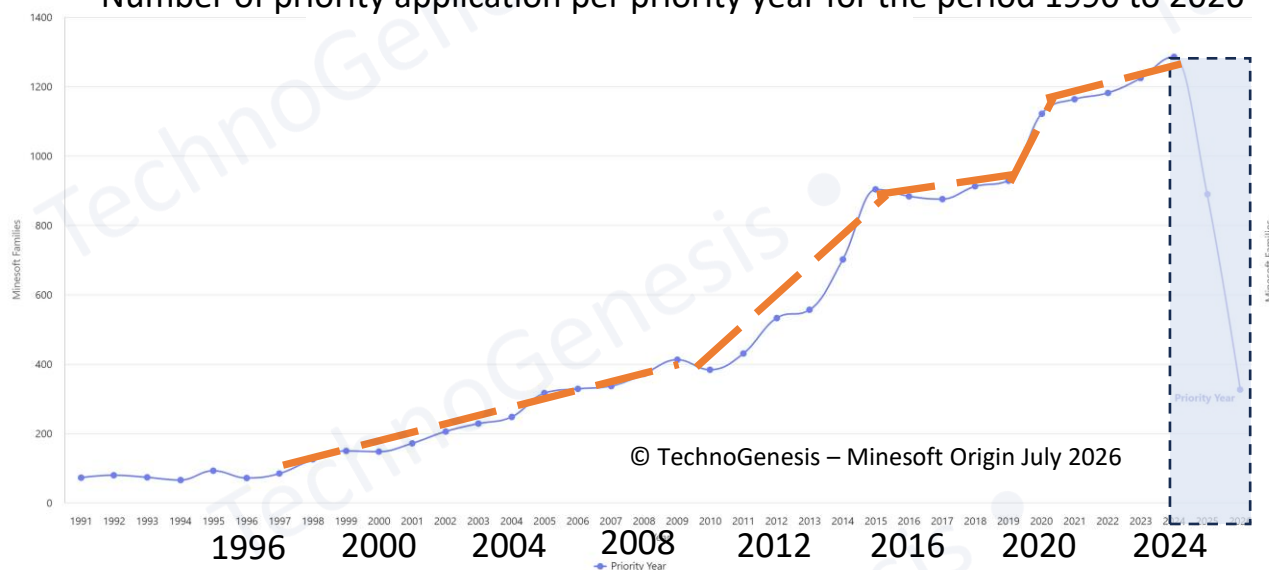
Carl Zeiss: Early & Sustained leader
Topcon & Alcon: Current front-runners
Canon: Optopol-driven wave (peak 2014, then retreat)
Nidek: Steady ophthalmic specialist.

Top 33 Jurisdictions by Minesoft Families



The US dominates, over the 2nd China by more than 60%, followed closely by 3rd Japan.

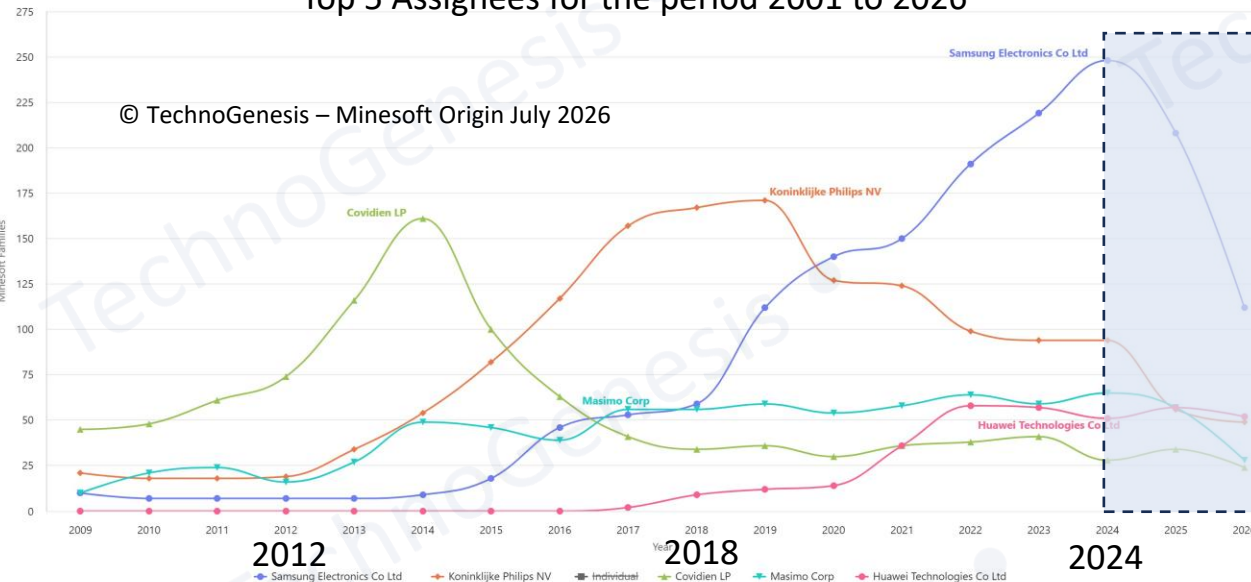
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Double S-curve: a first wave with the smartwatch / fitness-tracker (~2014–15), & then a second acceleration from 2020 (health monitoring) which is attributed to the COVID era.

Top 5 Assignees for the period 2001 to 2026

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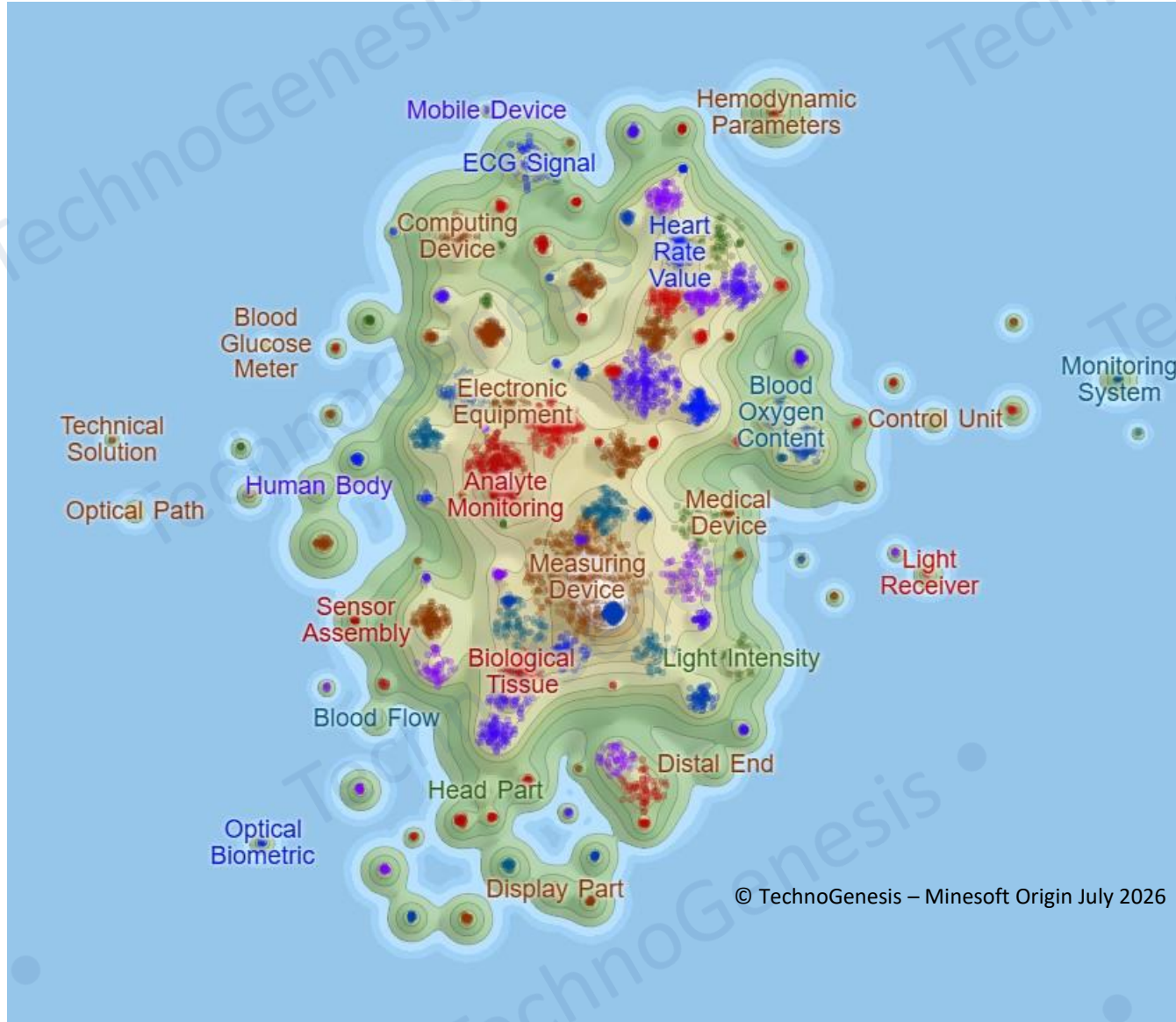
Samsung Electronics: Dominant and still rising to ~248 in 2024

Koninklijke Philips: peaked in 2019

Covidien LP: Early leader peaked in 2014

Masimo Corp: Pulse-oximetry pure player

Huawei: China entry in 2016



Fences (crowded, FTO risk) -> the vital-signs core — heart rate, ECG, blood-oxygen, hemodynamic parameters— plus the generic hardware massif (measuring device, sensor assembly, analyte monitoring)

White spaces (thin, at the edges): the optical-component periphery — optical path, light receiver, light intensity, optical biometric, blood flow — and the valleys between the consumer-device clusters (e.g. mobile/computing device, display part) and the clinical clusters.

Caution: empty doesn't mean valuable — a blank region can be infeasible rather than open.

S-curves, white spaces studies

Methodology: Patent families by earliest priority year, bounded by IPC/CPC codes and refined with keywords.

Two domains studied in-house with Minesoft Origin (hollow-core fibre; OCT & wearable biosensing); six read from published landscapes.

S-curves place a technology on its diffusion path —emergence, invasion, saturation — and the **theme map** locates the fences and the white spaces. Limits kept in view throughout: the **18-month black box**, and **publication ≠ granted rights**.

Results: Three cross-cutting signals

China leads filing counts; the US and Europe still hold the international families.

AI is the tide. Co-packaged optics, photonic computing and hollow-core fibre are one story — the physical layer of AI compute.

The domains are converging. Metasurfaces feed LiDAR and AR; silicon photonics underpins quantum and computing. **Whitespace** is found at the interfaces.

Next

What this analysis does *not* yet tell us: legal status (are the fences still live?), claim-level scope, and citation quality.

The next layer is assignee harmonisation, legal-status filtering and claim-level whitespace mapping which can turn a landscape into a filing strategy.